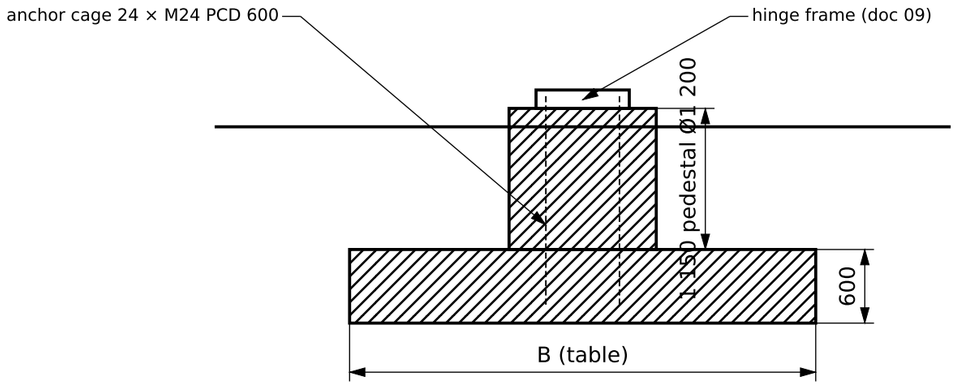
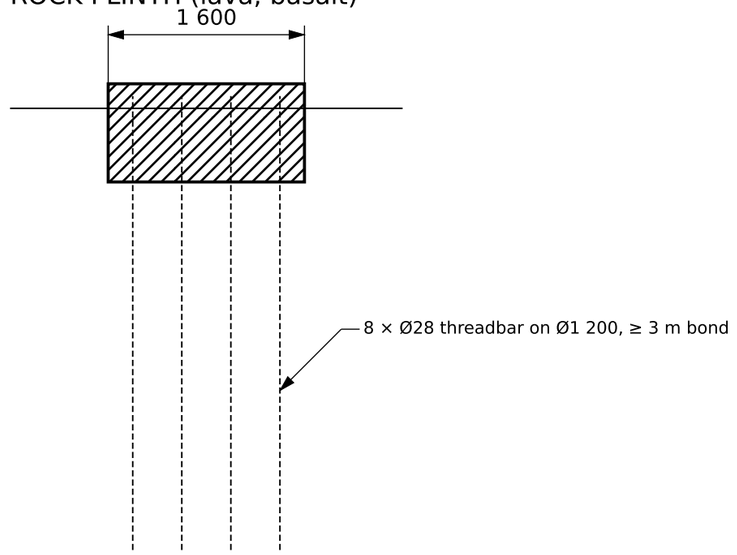
SPREAD FOOTING, SECTION



Ground	B m	Concrete m³	Bearing kPa	q_rd kPa
firm soil (dense gravel, moraine)	3.8	9.5	166.5	300.0
soft volcanic soil (loose ash, pumice sand)	4.2	11.4	95.9	100.0
frost-susceptible silt (thawed)	3.9	10.0	138.2	150.0
rock (anchored plinth)	1.6	2.0	–	anchors 249 kN

ROCK PLINTH (lava, basalt)



NOTES

1. Base 1.6 m below ground (frost depth). Design for the parked V_e50 base moment 617 kNm (doc 30 §6); characteristic eccentricity ≤ B/4, operation ≤ B/6.
2. Concrete C30/37 XF4 with air entrainment, cover 50 mm; rebar mesh top and bottom Ø16/150 both ways (to be detailed with the local engineer).
3. Rock: drill Ø76, grout cement 100 %, proof-load each anchor to 1.25 × design before tensioning; plinth cast on cleaned rock with 20 mm shear keys.
4. Soil class must be confirmed by a site inspection; soft volcanic ash needs a plate-load test.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FOUNDATION · SPREAD FOOTING AND ROCK PLINTH

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-500	A · 2026-11-06	1:50
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
C30/37 XF4 air-entrained; B500C rebar	9.5–11.4 m³ (soil), 2.0 m³ (rock)	Claude / not checked